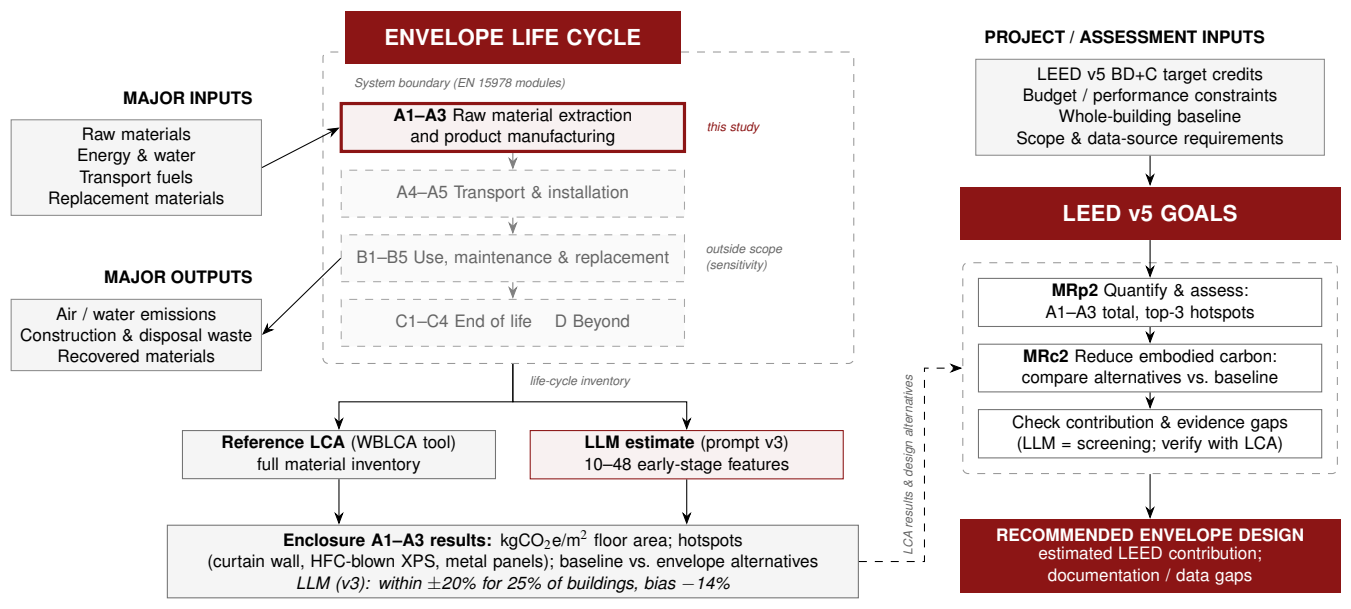
ENVELOPE LIFE CYCLE
System boundary (EN 15978 modules)
MAJOR INPUTS
Raw materials
Energy & water
Transport fuels
Replacement materials
A1–A3 Raw material extraction and product manufacturing
this study
A4–A5 Transport & installation
B1–B5 Use, maintenance & replacement
outside scope (sensitivity)
C1–C4 End of life D Beyond
MAJOR OUTPUTS
Air / water emissions
Construction & disposal waste
Recovered materials
life-cycle inventory
Reference LCA (WBLCA tool) full material inventory
LLM estimate (prompt v3) 10–48 early-stage features
Enclosure A1–A3 results: kgCO2e/m2 floor area; hotspots
(curtain wall, HFC-blown XPS, metal panels); baseline vs. envelope alternatives
LLM (v3): within ±20% for 25% of buildings, bias −14%
LCA results & design alternatives
PROJECT / ASSESSMENT INPUTS
LEED v5 BD+C target credits
Budget / performance constraints
Whole-building baseline
Scope & data-source requirements
LEED v5 GOALS
MRp2 Quantify & assess: A1–A3 total, top-3 hotspots
MRc2 Reduce embodied carbon: compare alternatives vs. baseline
Check contribution & evidence gaps (LLM = screening; verify with LCA)
RECOMMENDED ENVELOPE DESIGN
estimated LEED contribution; documentation / data gaps